

OpenSDI-based Deepfake Detection and Localization for the IJCAI 2026 DDL-X Challenge Track 3

Zhekai Wang and Tong Qiao*

School of Cyberspace, Hangzhou Dianzi University, Hangzhou, Zhejiang, China

Abstract

This paper describes our solution to Track 3 of the *IJCAI 2026 DDL 2.0 Workshop's DDL-X Challenge*, which evaluates deepfake image classification, manipulation localization and textual explanation generation simultaneously. Built on OpenSDI, a unified framework for open-world diffusion image detection and localization, our method adapts its MaskCLIP architecture via the Synergistic Pretrained Model (SPM) framework integrating CLIP's semantic comprehension and MAE's fine-grained spatial representation. We restructure the inference and submission pipeline for the challenge protocol: OpenSDI's pixel forgery masks are converted to bounding boxes with coordinate normalization mapped to [1, 1000], and outputs are formatted into structured JSON with classification, localization and explanation entries. We focus on the detection and localization objectives in this work and reserve the integration of explanation generation modules for future research. Even with this limitation, our method ranked 5th on the official DDL-X Track 3 leaderboard, proving pre-trained vision-language models' powerful generalization for joint deepfake detection and localization.

1 Introduction

Recent advances in artificial intelligence generated content (AIGC), particularly diffusion models and large multimodal models, have significantly improved the realism and accessibility of image manipulation techniques. While these technologies facilitate creative applications, they also increase the risk of malicious misuse, including misinformation dissemination, identity fraud, and social manipulation.

Traditional deepfake detection methods mainly focus on binary classification, determining whether an image is authentic or manipulated. However, such predictions often lack interpretability and fail to provide sufficient evidence for forensic analysis. Consequently, recent research has shifted toward deepfake detection, localization, and explainability,

aiming to provide not only authenticity judgments but also precise manipulation regions and human-understandable explanations.

To advance this direction, the IJCAI 2026 DDL 2.0 Workshop organizes the DDL-X Challenge Track 3, which jointly evaluates classification accuracy, localization quality, and textual interpretability on the DDL dataset [Miao *et al.*, 2025a]. The challenge adopts a comprehensive evaluation metric that combines classification accuracy (ACC), localization intersection-over-union (IoU), and explanation BERTScore [Zhang *et al.*, 2019].

Considering the diverse manipulation scenarios in DDL-X, we adopt OpenSDI as the backbone framework because of its open-world design for diffusion-generated image detection. OpenSDI¹ [Wang *et al.*, 2025] formulates diffusion-generated image detection as an open-world problem and introduces MaskCLIP, which synergizes pretrained CLIP and MAE models through prompt tuning and cross-attention mechanisms.

Although OpenSDI originally outputs image-level predictions and pixel-level forgery masks, the DDL-X benchmark requires structured JSON files containing bounding boxes and textual explanations. Therefore, we redesign the prediction pipeline to bridge the gap between OpenSDI outputs and challenge requirements.

Our contributions are summarized as follows:

- We adapt the OpenSDI framework to the DDL-X Challenge Track 3 by leveraging the MaskCLIP architecture for unified image-level deepfake detection and pixel-level forgery localization.
- We design a lightweight challenge-specific adaptation pipeline that converts pixel-level manipulation masks into benchmark-compatible bounding boxes with coordinate normalization, and automatically generates JSON submissions without introducing additional trainable parameters.
- Our method achieves the 5th place on the official DDL-X Track 3 leaderboard, demonstrating that the proposed challenge-oriented adaptation effectively transfers OpenSDI to the DDL-X benchmark. The explanation details will be extended in the future work.

*Corresponding author

¹<https://github.com/iamwangyabin/OpenSDI>

2 Related Work

2.1 Deepfake Detection and Localization

Early deepfake detection methods primarily formulated forgery analysis as a binary classification problem, aiming to distinguish authentic images from manipulated ones. To facilitate the development and evaluation of these methods, increasingly diverse benchmarks have been proposed, such as FaceForensics++ [Rossler *et al.*, 2019], Celeb-DF [Li *et al.*, 2020], and MFFI [Miao *et al.*, 2025b], which provide manipulated images under realistic scenarios for training and evaluation. Existing approaches typically exploit visual artifacts introduced during the image synthesis process, including frequency-domain inconsistencies [Jeong *et al.*, 2022; Tan *et al.*, 2024] and spatial-frequency fusion representations [Duan *et al.*, 2025; Wang *et al.*, 2023].

Convolutional neural networks (CNNs) have been widely adopted to capture local forgery traces through hierarchical feature extraction. Representative methods improve robustness by leveraging data augmentation [Wang and Deng, 2021], adversarial training [Chen *et al.*, 2022], and image reconstruction strategies [Cao *et al.*, 2022]. Meanwhile, transformer-based approaches [Masood *et al.*, 2023] have demonstrated superior capability in modeling long-range dependencies and global semantic inconsistencies, leading to significant improvements in cross-domain generalization.

Although image-level classification methods have achieved promising performance, they provide limited interpretability and fail to identify manipulated regions explicitly. To address this limitation, recent studies have extended deepfake detection to joint detection and localization tasks by introducing pixel-level supervision. Including MVSS-Net [Dong *et al.*, 2022], which combines multi-view feature learning with multi-scale supervision to capture both noise and boundary artifacts; OSN [Wu *et al.*, 2022], which improves robustness against lossy image processing; HiFi-Net [Guo *et al.*, 2023], which employs multi-branch feature extraction for synthetic image localization; and IML-ViT [Ma *et al.*, 2023], which integrates vision transformers with feature pyramid networks and edge-aware constraints to enhance localization performance.

More recently, pretrained vision-language models have demonstrated strong generalization capabilities for open-world deepfake detection. OpenSDI [Wang *et al.*, 2025] introduces MaskCLIP, a synergistic pretrained framework that combines the semantic understanding ability of CLIP with the fine-grained spatial representation capability of MAE through prompt tuning and cross-attention mechanisms. By jointly optimizing image-level detection and pixel-level localization, MaskCLIP achieves state-of-the-art performance across multiple unseen generative domains.

Motivated by its strong cross-domain generalization ability, we adopt OpenSDI as the backbone of our competition system and further adapt its outputs to satisfy the challenge-specific submission requirements.

2.2 Explainable Deepfake Detection

Despite substantial progress in deepfake detection and localization, most existing methods remain limited to binary de-

cisions or pixel-level predictions, offering little insight into the rationale behind their outputs. This lack of interpretability restricts their applicability in high-stakes scenarios such as digital forensics, content moderation, and misinformation mitigation.

The rapid development of large multimodal models has made this objective increasingly feasible. Representative models such as LISA [Lai *et al.*, 2024] demonstrate that integrating visual perception with language reasoning enables simultaneous segmentation and natural language understanding, providing an effective foundation for explainable image forensics.

Building upon these advances, several studies have explored multimodal frameworks for explainable forgery analysis. AntiFakePrompt [Chang *et al.*, 2023] formulates deepfake detection as a visual question answering task by optimizing soft prompts for vision-language models. ForgeryGPT [Liu *et al.*, 2024] incorporates forensic knowledge into a mask-aware multimodal framework to jointly perform forgery detection, localization, and explanation generation. FakeShield [Xu *et al.*, 2025] leverages multimodal large language models to identify manipulated regions while providing interpretable textual evidence for its predictions. SIDA [Huang *et al.*, 2025] introduces a unified framework for social media image deepfake detection, localization, and explanation. By integrating visual features with language models, SIDA generates pixel-level forgery masks and natural language explanations simultaneously, significantly improving model transparency.

The DDL-X Challenge explicitly evaluates explainability alongside detection and localization, reflecting the growing demand for transparent deepfake forensic systems. In this work, we focus on the detection and localization objectives by adapting OpenSDI to the challenge setting. The integration of explanation generation modules is reserved for future work.

3 Method

3.1 Overall Framework

Our framework basically follows the original OpenSDI MaskCLIP architecture. Given an input image, the CLIP visual encoder extracts high-level semantic representations for image-level classification, while the MAE encoder captures fine-grained spatial features for forgery localization. Visual cross-attention modules align CLIP and MAE features at multiple scales, and textual-visual cross-attention modules integrate class-specific prompts into the localization decoder. In this context, the final outputs include:

- Image-level authenticity prediction score;
- Pixel-level forgery probability map.

3.2 Challenge-specific Adaptation

The original OpenSDI framework is designed for image-level authenticity classification and pixel-level manipulation localization. In contrast, the DDL-X benchmark formulates deepfake analysis as a multi-task understanding problem, requiring each prediction to be represented as a structured JSON file containing authenticity labels, manipulation regions, and textual explanations.

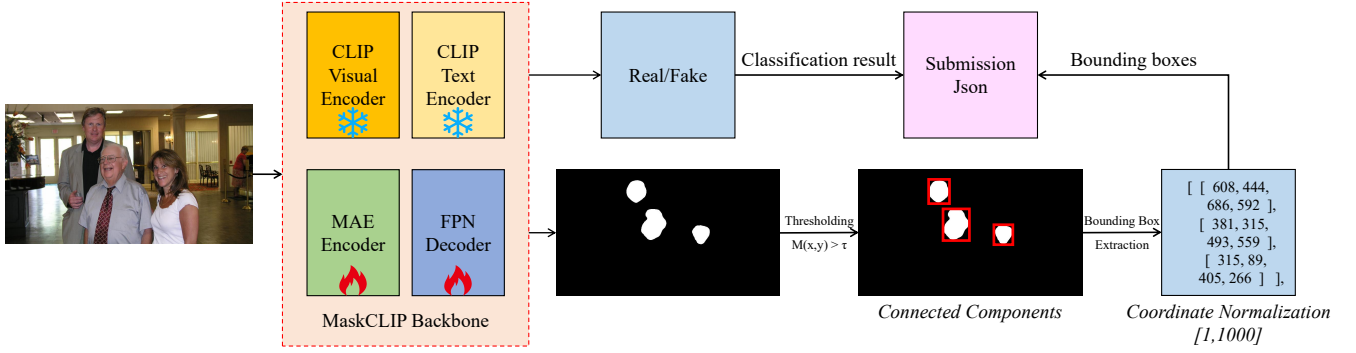


Figure 1: Overview of our OpenSDI-based framework for the DDL-X Challenge Track 3. The input image is first processed by the MaskCLIP backbone, which jointly performs image-level authenticity classification and pixel-level forgery localization. The predicted forgery probability map is subsequently converted into challenge-compliant manipulation regions through thresholding, connected component analysis, and bounding box extraction. Finally, the detected regions are normalized to the coordinate range $[1, 1000]$ and combined with the classification result to generate the required JSON submission file. Frozen modules are marked with snowflakes, while trainable modules are marked with flames.

To bridge the gap between the original model outputs and the challenge requirements, we introduce a lightweight challenge-specific adaptation pipeline that transforms segmentation-oriented predictions into a unified representation compatible with the evaluation protocol. Notably, the proposed adaptation introduces no additional trainable parameters and preserves the original training objectives.

Image-level Classification

MaskCLIP jointly predicts an image-level authenticity label and a manipulation mask. We directly employ the classification branch to determine whether an input image is authentic or manipulated. The predicted labels are converted into textual outputs, where “real” and “fake” denote authentic and manipulated images, respectively.

Manipulation Region Localization

The localization branch generates a dense manipulation probability map $M(x, y)$. A fixed threshold τ is applied to obtain a binary manipulation mask:

$$M_b(x, y) = \begin{cases} 1, & M(x, y) > \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Connected component analysis is then performed on the binary mask to identify individual manipulated regions. For each connected component, an axis-aligned bounding box (x_1, y_1, x_2, y_2) is generated. Small isolated regions are further removed using an area-based filtering strategy to suppress noisy predictions.

Following the challenge specification, all bounding box coordinates are normalized to the range $[1, 1000]$ according to the original image dimensions:

$$\begin{aligned} x'_1 &= \text{round} \left(\frac{x_1}{W} \times 1000 \right), \\ y'_1 &= \text{round} \left(\frac{y_1}{H} \times 1000 \right), \\ x'_2 &= \text{round} \left(\frac{x_2}{W} \times 1000 \right), \\ y'_2 &= \text{round} \left(\frac{y_2}{H} \times 1000 \right). \end{aligned} \quad (2)$$

where W and H denote the width and height of the original image, respectively. For images classified as authentic, the bounding box field is set to `null`.

Interpretability Output

Besides image classification and manipulation localization, the DDL-X benchmark also requires a natural language description of the visible forgery traces. While OpenSDI benefits from vision-language representations inherited from CLIP, generating such explanations requires additional image-text supervision.

Under the competition protocol, only the official DDL dataset is permitted for training. Since this dataset does not provide textual explanation annotations, it is infeasible to train or fine-tune an explanation generation module without introducing external supervision, which is prohibited by the challenge rules.

Therefore, our submission concentrates on the two primary objectives that can be fully optimized using the available supervision, namely image-level forgery detection and pixel-level manipulation localization. The explanation field is intentionally left empty to comply with the competition protocol, while the proposed framework naturally supports future integration of explanation generation modules when appropriate supervision is available.

Competition Results

Figure 2 presents the final leaderboard released by the official DDL-X Challenge organizers. Our team “newtester” achieved 5th place in the final ranking. The final leaderboard was determined according to the official evaluation protocol, which combines the public and private results. These results demonstrate the effectiveness of the proposed challenge-oriented adaptation under the DDL-X evaluation protocol.

4 Experiments

4.1 Training and Inference Configuration

To adapt the original OpenSDI framework to the DDL-X challenge, we fine-tuned the MaskCLIP model using the offi-

Track 3: Deepfake Detection, Localization, and Explainability

Rank	Team	Affiliation	Final Score
1	HIT_VIRLAB	Harbin Institute of Technology	0.8940
2	CBSR_what_the_dog_doing	University of Chinese Academy of Sciences	0.7025
3	XQ999	Independent Researcher	0.5770
4	ACVLAB	National Yang Ming Chiao Tung University (ACVLab)	0.3505
5	newtester	Hangzhou Dianzi University	0.2522
6	Institute of Image Processing and Pattern Recognition, Hebei University of Technology	School of Artificial Intelligence, Hebei University of Technology	0.1630

Figure 2: Public leaderboard of the IJCAI 2026 DDL-X Challenge Track 3. Our submission (“newtester”) ranked 5th.

Table 1: Settings of Training Hyperparameters

Parameter	Value
Backbone	MaskCLIP (ViT-L/14)
Input resolution	512×512
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.05
Batch size	8
Epochs	5
Warm-up epochs	0
Gradient accumulation	1
Number of workers	4
Random seed	42
Edge mask width	7
Sample ratio	0.01

cial training set under a single-GPU setting.

Training Setup

All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU with 24 GB memory. The model was implemented using the PyTorch deep learning framework.

Following the original OpenSDI implementation, we adopted the ViT-L configuration of MaskCLIP as the backbone network. During training, the input images were resized to 512×512 pixels.

To reduce computational cost and accelerate model adaptation, only 1% of the official training set was randomly sampled for fine-tuning. This strategy significantly reduced the training time while maintaining competitive performance on the validation set. The detailed training hyperparameters are summarized in Table 1.

Data augmentation techniques were applied during training to improve model robustness, including random scaling, horizontal and vertical flipping, random brightness and contrast adjustment, image compression, random rotation, and Gaussian blurring. The optimization objective combines three loss terms:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{bce} + \lambda \mathcal{L}_{edge}, \quad (3)$$

Table 2: F1 Score of Validation Pixel over Training Epochs

Epoch	F1 Score
0	0.5804
1	0.6284
2	0.6609
3	0.6637
4	0.6775

Table 3: Software Environment Versions

Component	Version
Operating system	Ubuntu Linux
Python	3.12
PyTorch	2.4.1
CUDA	12.1
NVIDIA Driver	595.71.05

where $\mathcal{L}_{ce}$ denotes the image-level classification loss, $\mathcal{L}_{bce}$ represents the pixel-level binary cross-entropy loss for manipulation localization, and $\mathcal{L}_{edge}$ is the edge-aware weighted loss that emphasizes boundary regions. The weighting coefficient λ is set to 20.

Validation Strategy

The official validation set was used for model selection. We evaluated the localization performance using pixel-level F1 score and saved the checkpoint with the best validation performance. The validation results across training epochs are shown in Table 2.

The model trained for five epochs achieved the best validation performance at Epoch 4, with a pixel-level F1 score of 0.6775. Therefore, the checkpoint from Epoch 4 was selected for final testing.

Inference Setup

During inference, the selected checkpoint was applied to all 100,000 test images provided by the challenge organizers. The test images were resized to 512×512 pixels and processed with a batch size of 32. Predictions from the classification branch and localization branch were converted into the required JSON submission format through the challenge-specific adaptation pipeline described in Section 3.2. No test-time augmentation, model ensembling, or additional post-training optimization techniques were employed.

Computational Efficiency

4.2 Implementation Details

The proposed framework was implemented based on the publicly available OpenSDI codebase and customized for the DDL-X challenge.

Software Environment

The experiments were carried out under the software environment detailed in Table 3.

The training and inference pipelines were developed using PyTorch and several open-source libraries, including:

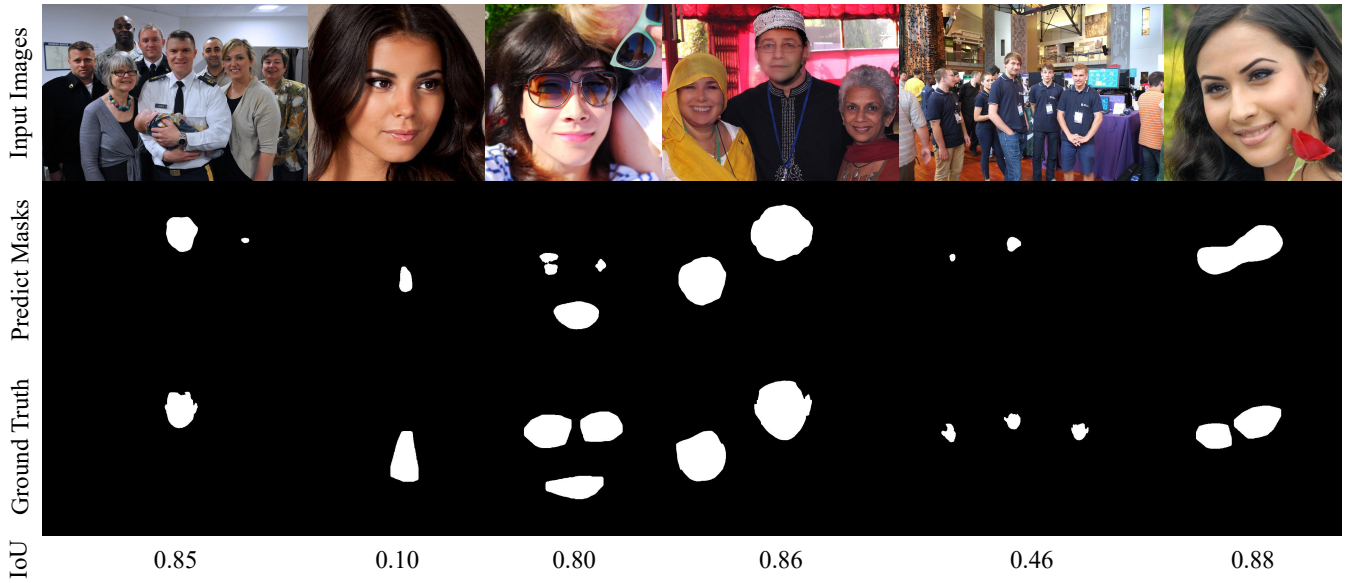


Figure 3: Qualitative comparison on the DDL-X Track 3 validation set.



Figure 4: Visualization results on the DDL-X Track 3 test set.

- OpenCV for image processing and connected component analysis;
- Albumentations for data augmentation;
- Weights & Biases and TensorBoard for experiment tracking and visualization.

Reproducibility

To ensure reproducibility, all experiments were conducted with a fixed random seed of 42. The training and inference procedures were executed using a single GPU without distributed training. The model weights corresponding to the highest validation pixel-level F1 score were used for final evaluation.

Training efficiency is an important consideration for practical deployment. By leveraging only 1% of the training data, the proposed framework substantially reduced computational overhead while preserving competitive localization performance. This lightweight fine-tuning strategy enables rapid

adaptation of large-scale segmentation-based deepfake detectors to challenge-specific evaluation protocols without requiring extensive computational resources. Furthermore, the challenge-specific adaptation module introduced in Section 3.2 is entirely post-processing based and introduces no additional trainable parameters. Consequently, the overall computational complexity during inference remains nearly identical to that of the original OpenSDI framework.

Submission

The complete inference pipeline consists of four sequential stages:

1. Manipulation probability prediction using MaskCLIP;
2. Image-level authenticity classification;
3. Connected component extraction and bounding box generation;
4. Automatic JSON file generation for challenge submission.

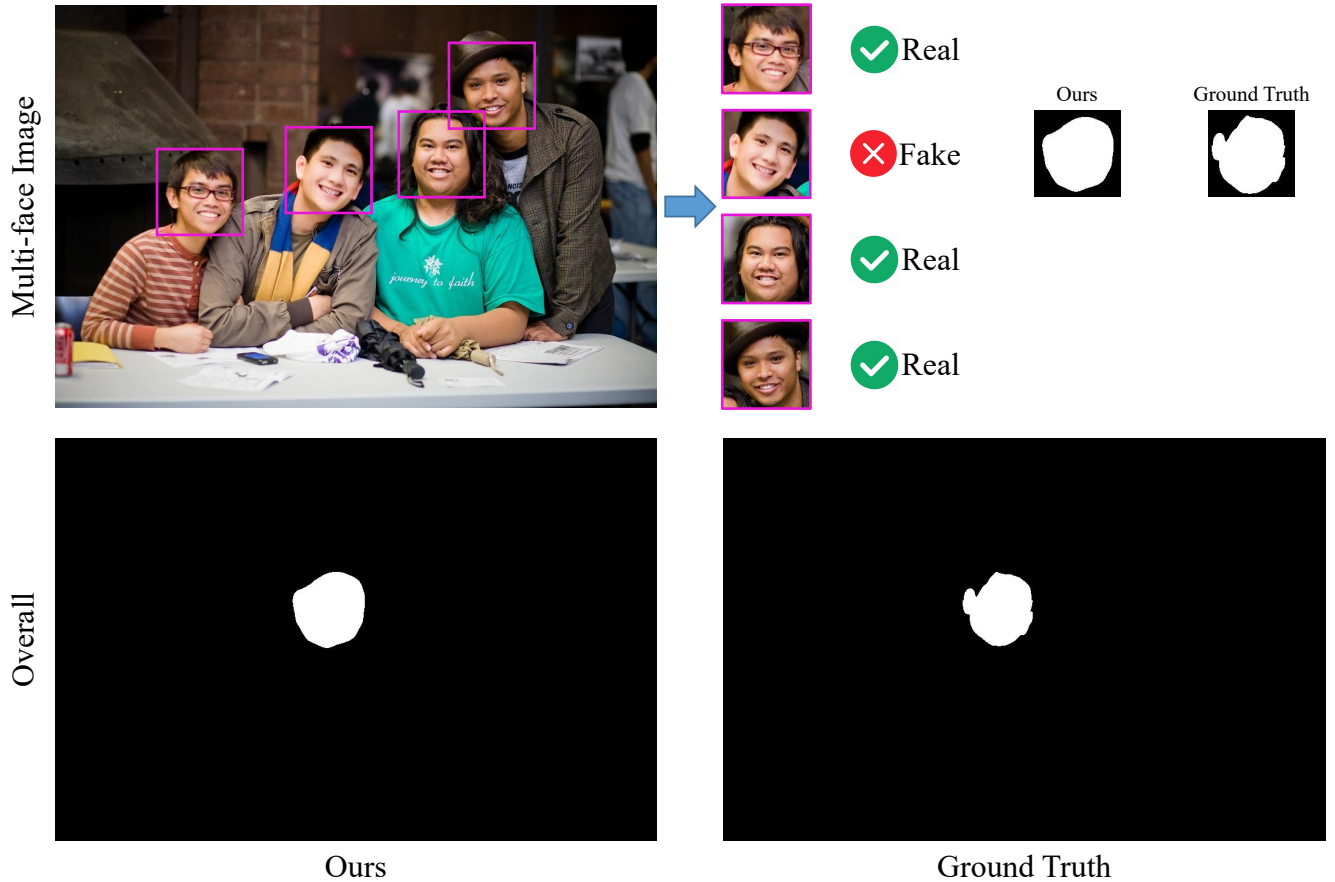


Figure 5: Visualization of a challenging multi-face forgery example from the DDL-X validation set. The proposed method first performs face-level authenticity analysis and successfully identifies the manipulated individual among several authentic faces.

The final outputs strictly follow the official challenge format, generating one JSON file for each test image. This design enables seamless integration of segmentation-based forgery detectors into structured deepfake understanding benchmarks.

Qualitative Results

To further evaluate the behavior of the proposed framework, we present qualitative results on both the DDL-X validation set and representative challenging samples.

Figures 3 and 4 visualize the predicted forgery masks produced by our method. On the validation set, the predicted masks generally exhibit strong spatial consistency with the corresponding ground-truth annotations. The model successfully identifies manipulated facial regions and accurately captures their approximate shapes and locations. Although minor discrepancies can be observed around object boundaries and extremely small manipulated areas, the overall localization quality remains consistently performs across the DDL-X validation set.

Figure 5 further presents a challenging multi-face example. Unlike conventional samples containing a single manipulated subject, this image includes multiple facial instances with only one forged identity. The proposed framework suc-

cessfully distinguishes the manipulated face from authentic individuals and generates a localization mask that closely matches the ground truth. This result demonstrates that the OpenSDI-based framework is capable of focusing on forgery-related artifacts rather than merely relying on facial presence, thereby maintaining reliable localization performance on representative multi-face samples in the DDL-X validation set.

Overall, the qualitative results confirm that our adaptation of OpenSDI provides accurate manipulation localization while preserving strong generalization ability under complex visual conditions.

5 Discussion

5.1 Conclusion

This paper presents our solution to the IJCAI 2026 DDL-X Challenge Track 3. Built upon the OpenSDI framework, our approach combines image-level authenticity classification with pixel-level manipulation localization through pretrained vision-language representations. To satisfy the DDL-X evaluation protocol, we further develop a lightweight post-processing pipeline that converts predicted manipulation masks into normalized bounding-box annotations and benchmark-compatible JSON outputs without introducing

additional trainable parameters. The proposed solution achieved 5th place on the official DDL-X Track 3 leaderboard, demonstrating the effectiveness of the proposed challenge-oriented adaptation on this benchmark.

5.2 Limitations and Future Work

Although the proposed framework achieves promising performance on the DDL-X benchmark, several limitations remain.

First, our method is adapted from the original OpenSDI framework and primarily focuses on challenge-oriented deployment rather than architectural innovation. Consequently, its performance is inherently bounded by the capability of the backbone model.

Second, the current system generates authenticity predictions and manipulation localization results only. Although the DDL-X benchmark also includes textual explanation as one of its tasks, the official training set does not provide supervision for explanation generation. Therefore, explanation generation is left for future work.

Finally, our evaluation is conducted under the DDL-X benchmark protocol. Future work will evaluate on additional benchmarks of the proposed adaptation pipeline and explore unified multimodal frameworks that jointly perform detection, localization, and explanation generation.

References

- [Cao *et al.*, 2022] Junyi Cao, Chao Ma, Taiping Yao, Shen Chen, Shouhong Ding, and Xiaokang Yang. End-to-end reconstruction-classification learning for face forgery detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4113–4122, 2022.
- [Chang *et al.*, 2023] You-Ming Chang, Chen Yeh, Wei-Chen Chiu, and Ning Yu. Antifakeprompt: Prompt-tuned vision-language models are fake image detectors. *arXiv preprint arXiv:2310.17419*, 2023.
- [Chen *et al.*, 2022] Liang Chen, Yong Zhang, Yibing Song, Lingqiao Liu, and Jue Wang. Self-supervised learning of adversarial example: Towards good generalizations for deepfake detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 18710–18719, 2022.
- [Dong *et al.*, 2022] Chengbo Dong, Xinru Chen, Ruohan Hu, Juan Cao, and Xirong Li. Mvss-net: Multi-view multi-scale supervised networks for image manipulation detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3):3539–3553, 2022.
- [Duan *et al.*, 2025] Junxian Duan, Yuang Ai, Jipeng Liu, Shenyuan Huang, Huaibo Huang, Jie Cao, and Ran He. Test-time forgery detection with spatial-frequency prompt learning. *International Journal of Computer Vision*, 133(2):672–687, 2025.
- [Guo *et al.*, 2023] Xiao Guo, Xiaohong Liu, Zhiyuan Ren, Steven Grosz, Iacopo Masi, and Xiaoming Liu. Hierarchical fine-grained image forgery detection and localization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3155–3165, 2023.
- [Huang *et al.*, 2025] Zhenglin Huang, Jinwei Hu, Xiangtai Li, Yiwei He, Xingyu Zhao, Bei Peng, Baoyuan Wu, Xiaowei Huang, and Guangliang Cheng. Sida: Social media image deepfake detection, localization and explanation with large multimodal model. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 28831–28841, 2025.
- [Jeong *et al.*, 2022] Yonghyun Jeong, Doyeon Kim, Youngmin Ro, and Jongwon Choi. Fregan: robust deepfake detection using frequency-level perturbations. In *Proceedings of the AAAI conference on artificial intelligence*, volume 36, pages 1060–1068, 2022.
- [Lai *et al.*, 2024] Xin Lai, Zhuotao Tian, Yukang Chen, Yanwei Li, Yuhui Yuan, Shu Liu, and Jiaya Jia. Lisa: Reasoning segmentation via large language model. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9579–9589, 2024.
- [Li *et al.*, 2020] Yuezun Li, Xin Yang, Pu Sun, Honggang Qi, and Siwei Lyu. Celeb-df: A large-scale challenging dataset for deepfake forensics. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3207–3216, 2020.
- [Liu *et al.*, 2024] Jiawei Liu, Fanrui Zhang, Jiaying Zhu, Esther Sun, Qiang Zhang, and Zheng-Jun Zha. Forgerygpt: Multimodal large language model for explainable image forgery detection and localization. *arXiv preprint arXiv:2410.10238*, 2024.
- [Ma *et al.*, 2023] Xiaochen Ma, Bo Du, Zhuohang Jiang, Xia Du, Ahmed Y Al Hammadi, and Jizhe Zhou. Iml-vit: Benchmarking image manipulation localization by vision transformer. *arXiv preprint arXiv:2307.14863*, 2023.
- [Masood *et al.*, 2023] Momina Masood, Mariam Nawaz, Khalid Mahmood Malik, Ali Javed, Aun Irtaza, and Hafiz Malik. Deepfakes generation and detection: state-of-the-art, open challenges, countermeasures, and way forward: Deepfakes generation and detection: state-of-the-art, open challenges, countermeasures, and way forward. *Applied intelligence*, 53(4):3974–4026, 2023.
- [Miao *et al.*, 2025a] Changtao Miao, Yi Zhang, Weize Gao, Zhiya Tan, Weiwei Feng, Man Luo, Jianshu Li, Ajian Liu, Yunfeng Diao, Qi Chu, et al. Ddl: A large-scale datasets for deepfake detection and localization in diversified real-world scenarios. *arXiv preprint arXiv:2506.23292*, 2025.
- [Miao *et al.*, 2025b] Changtao Miao, Yi Zhang, Man Luo, Weiwei Feng, Kaiyuan Zheng, Qi Chu, Tao Gong, Jianshu Li, Yunfeng Diao, Wei Zhou, et al. Mffi: Multi-dimensional face forgery image dataset for real-world scenarios. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 13235–13242, 2025.
- [Rossler *et al.*, 2019] Andreas Rossler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. Faceforensics++: Learning to detect manipulated facial images. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1–11, 2019.

- [Tan *et al.*, 2024] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. Frequency-aware deepfake detection: Improving generalizability through frequency space domain learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 5052–5060, 2024.
- [Wang and Deng, 2021] Chengrui Wang and Weihong Deng. Representative forgery mining for fake face detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 14923–14932, 2021.
- [Wang *et al.*, 2023] Yuan Wang, Kun Yu, Chen Chen, Xiyuan Hu, and Silong Peng. Dynamic graph learning with content-guided spatial-frequency relation reasoning for deepfake detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 7278–7287, 2023.
- [Wang *et al.*, 2025] Yabin Wang, Zhiwu Huang, and Xiaopeng Hong. Opensdi: Spotting diffusion-generated images in the open world. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4291–4301, 2025.
- [Wu *et al.*, 2022] Haiwei Wu, Jiantao Zhou, Jinyu Tian, and Jun Liu. Robust image forgery detection over online social network shared images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13440–13449, 2022.
- [Xu *et al.*, 2025] Zhipei Xu, Xuanyu Zhang, Runyi Li, Zecheng Tang, Qing Huang, and Jian Zhang. Fakeshield: Explainable image forgery detection and localization via multi-modal large language models. In *International Conference on Learning Representations*, volume 2025, pages 31186–31216, 2025.
- [Zhang *et al.*, 2019] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*, 2019.